

Project Scope Statement

Mumbai–Ahmedabad High-Speed Rail Corridor – Proposal Development for Efficient Timetable Planning and Execution

1. Project Objectives

The primary objective of this project is to develop a **comprehensive, cost-effective, and data-driven proposal** for the planning and execution of train schedules along the Mumbai–Ahmedabad High-Speed Rail Corridor. This proposal will aim to:

- **Enhance efficiency and punctuality** of train operations using AI/ML-based predictive analytics.
- **Ensure compliance** with NHRCL guidelines and international railway standards.
- **Optimize resource allocation** to improve operational efficiency and minimize delays.
- **Deliver a high-quality proposal** that includes technical feasibility analysis, financial projections, and an implementation roadmap for NHRCL.

2. Project Scope Description

The project involves the development of a detailed **timetable planning and execution proposal** that incorporates:

- **Automated scheduling models** that optimize train operations based on passenger demand, station capacity, and energy efficiency.
- **Real-time monitoring and adjustment mechanisms** for delay management and dynamic schedule updates.
- **Stakeholder collaboration** with NHRCL and regulatory bodies to align the proposal with operational needs.
- **Financial analysis** including cost estimates, return on investment (ROI), and funding requirements.
- **Regulatory compliance framework** to ensure adherence to Indian Railways' safety and operational guidelines.

In-Scope:

- AI/ML-driven timetable modeling and optimization.
- Compliance assessment against Indian and international railway standards.
- Proposal documentation, including technical details and cost estimates.
- Development of presentation materials for NHRCL stakeholders.

Out-of-Scope:

- Direct implementation of the scheduling system.
- Development of hardware infrastructure for railway operations.
- Deployment of AI/ML models in live railway operations.

3. Project Deliverables

The key deliverables for this project include:

- **Proposal Document:**
 - Executive summary and strategic objectives.
 - Detailed project scope, methodology, and implementation roadmap.
 - Cost-benefit analysis and ROI projections.
- **Technical Solution:**
 - AI/ML-driven timetable optimization framework.
 - System architecture for real-time monitoring.
 - Simulation models for schedule feasibility validation.
- **Presentation Materials:**
 - PowerPoint presentations for NHSRCL stakeholders.
 - Supporting documents, graphs, and data visualizations.

4. Acceptance Criteria

For the proposal and deliverables to be accepted, the following conditions must be met:

- The proposal aligns with NHSRCL's strategic goals and operational requirements.
- AI-driven scheduling models achieve at least **95% accuracy** in simulations.
- Compliance with NHSRCL and **international high-speed rail standards (UIC, ISO 9001:2015)**.
- **Stakeholder approval** from NHSRCL's review committee.

5. Project Constraints

- **Time:** The final proposal must be submitted by **June 15, 2025**.
- **Budget:** Total budget is estimated at **€950,000**, with allocated costs for personnel, R&D, and documentation.
- **Regulatory Compliance:** The solution must meet **Indian Railways' safety and operational regulations**.
- **Technology:** AI/ML models must be compatible with NHSRCL's existing IT infrastructure.

6. Project Assumptions

- NHSRCL will provide **timely access to operational data** required for timetable modeling.
- The project team will have access to **sufficient computational resources** for AI/ML simulations.
- Key stakeholders will be available for **collaboration and review meetings** as scheduled.
- The **budget and funding** for proposal development will remain unchanged throughout the project duration.

7. Project Exclusions

The project does not include:

- Physical construction or modification of railway infrastructure.
- Deployment of scheduling models into NHSRCL’s operational system.
- Real-time AI testing with actual train movements.

8. Project Boundaries

The project will start with a **requirement analysis phase** (Feb 2025) and conclude with the **submission of the final proposal (June 2025)**. Key boundary conditions include:

- **Geographical scope:** The Mumbai–Ahmedabad High-Speed Rail Corridor.
- **Operational focus:** Timetable planning and execution, not railway maintenance or infrastructure development.

9. Project Organization

Project Team Roles and Responsibilities

- **Project Sponsor (Nicole Schelter, CEO, via Teams):** Provides executive oversight and approvals.
- **Project Manager (Roger Candoo, Teamlead Timetable Planning and Execution, via Teams):** Leads project execution, resource management, and stakeholder coordination.
- **Technical Team:** Develops AI/ML models and timetable simulations.
- **Quality Assurance Team:** Ensures compliance with regulatory and performance standards.
- **Stakeholder Representatives (NHSRCL officials):** Review and validate project deliverables.

10. Project Risks

Identified Risks and Mitigation Strategies

Risk	Potential Impact	Mitigation Strategy
Requirement misalignment	Project delays due to unclear NHSRCL expectations	Frequent stakeholder communication and validation checkpoints
Budget constraints	Limited funding affecting project scope	Provide cost-efficient alternative solutions
AI/ML model performance issues	Inaccurate scheduling predictions	Continuous testing and validation using historical data
Regulatory compliance issues	Delays due to non-compliance	Engage compliance experts early in the project
NHSRCL approval delays	Extended project timeline	Conduct weekly progress updates with NHSRCL decision-makers